Chapter 1: Entering the World of Secondary Science

Quick revision notes | NCERT Grade 9 Science: *Exploration*

Big idea in one line

Science is not only **what** we know but **how** we know it: observe → measure → model → express with symbols/maths → predict → test → revise.

1. Models: simplifying reality

- The natural world is too complex to study in full detail, so science uses **models**: simplified versions of a system that keep only what matters for the question asked.
- Building a model = **making assumptions + deliberately ignoring some details**. This is done on purpose; it is **not a mistake**.

Subject	How the model simplifies
Physics	Moving car treated as a single point; air resistance ignored for a falling object
Chemistry	Atoms and molecules drawn as spheres and bonds
Biology	Cell shown as a diagram of key parts; individual cells ignored when studying the heart as a pump
Earth science	Earth treated as a smooth sphere with distinct layers

Example 1.1: Cricket six

Key question: Will the ball cross the boundary without touching the ground?

Keep: mass of ball, speed, direction of hit. **Small effect (ignore at first):** air resistance, spin, seam stitching. **Irrelevant:** bat brand, ball colour, grass.

Better models add more details later for greater accuracy.

Meet a Scientist: Meghnad Saha

Treated star matter as a **hot gas**, focusing only on temperature, pressure and ionisation of atoms. This explained how a star's **colour is linked to its temperature**.

2. Language of science: terms, symbols, units

- Everyday words (*force, work, cell, reaction*) have **precise scientific meanings**, so ideas can be communicated without ambiguity worldwide.
- Quantities have **symbols** and defined **units**: mass **m**, velocity **v**, force **F**, electric current **I**.
- Symbols come from history and international agreement. Speed of light **c** is from Latin *celeritas* (speed); it is defined as exactly **299 792 458 m/s**.
- **Standard (SI) units** make results comparable and trade fair. Example: a kilogram means the same everywhere.

Case study: why units matter

A passenger aircraft ran out of fuel mid-flight because fuel density was used in **pounds** per litre instead of **kilograms** per litre. It needed 22,300 kg but was about 15,000 litres short. It glided to an emergency landing (aircraft damaged, no casualties). **Lesson: use SI units consistently.**

3. Mathematics: a language for thinking

- An equation is a **compact statement of how quantities are related**, not just a calculation tool.
- Used for motion (distance, time, velocity), rates of chemical reactions, population growth, energy changes.
- **Method:** understand the situation → identify relevant quantities → apply relationships to reason. Understanding matters more than memorising formulas.

4. Laws, theories, principles

Term	Meaning	Example
Law	Describes a **regular pattern** seen in nature (words or maths)	Newton's laws of motion explain the jerk when a bus stops suddenly
Theory	**Explains why** the pattern occurs, based on tested evidence	Atomic theory explains how molecules form
Principle	A **broad idea** that helps make sense of a situation	Conservation of energy while climbing stairs

- A scientific theory is **NOT a guess**. It is a well-tested explanation that stays open to improvement as new evidence arrives, which is what makes science reliable.

5. Prediction and testing

- Established laws, theories and models let us **predict** what will happen in new conditions, even when we cannot experiment (e.g., how far a kicked football travels, CO_2 produced in a reaction, change in breathing while running).
- Predictions are **reasoned expectations from evidence**, not guesses.
- **Match** with observation → confidence grows. **Mismatch** → re-examine assumptions, models or measurements, which drives further exploration.
- Even the best theories have **limits** and may fail in new conditions or with more precise measurements. Such failure is science's **greatest strength**: ideas are rejected only on **evidence**, never opinion or belief. No theory is final.

Example 1.2: Making "It will rain because clouds look dark" testable

Ask for **measurable evidence and past patterns**, not yes/no answers: sky condition at last rain, today's humidity (was it above 80% last time?), wind speed and direction, is temperature falling as before?

Why forecasts go wrong

Weather depends on temperature, pressure, humidity and wind. Tiny differences in starting conditions **grow over time**, so forecasts are reliable for hours to a few days but less certain beyond.

Checking viral claims: food during an eclipse

Claim: food becomes harmful during an eclipse. Test with questions: an eclipse is just a **play of shadows**; no major temperature change; food does not spoil just because it is in shadow. **No physical, chemical or biological mechanism supports the claim.**

6. Estimation: reasoning over exact calculation

- Strategy: **understand the situation** → **identify key quantities** → **make a rough estimate** to check if an answer is sensible.
- Estimation builds intuition, helps **detect errors** and shows if a result is reasonable or impossible.

Example 1.3: Litres of air breathed per day

Breaths: 12–15 per min × 1440 min ≈ **20,000 breaths/day**. One breath: ~4–5 breaths fill a 2 L balloon, so ≈ **0.5 L**. Total ≈ 20,000 × 0.5 = **~10,000 L/day**.

Cross-check: 3 balloons/min × 2 L × 1440 min = **8,640 L**, close to 10,000 L, so the estimate is reasonable.

Rice estimate (Ready to Go Beyond)

Assume all calories come from rice. Adult needs ~2000–2500 kcal/day. Find kcal in 100 g rice, then scale to family of 4 for a month. Sanity check: 100 g/month is far too little, a few tonnes is far too much.

7. Connections between branches of science

- After Grade 10, science splits into **physics, chemistry, biology, earth science**. These divisions are human-made to organise knowledge; **nature has no such boundaries**.

- Real problems (climate change, medicines, sustainable technology) need **several disciplines together**. Science also links with mathematics, technology, arts and social sciences.
- **Example, a mask:** physics (particle motion, electrostatic attraction), chemistry (polymer fibres), biology (size and behaviour of viruses), mathematics (airflow and filtration efficiency).

8. Science as a human activity

- Shaped by **curiosity, creativity, collaboration and careful questioning**; grows as people ask, test, share results and learn from mistakes.
- Developed over time by many people across cultures and generations.
- Scientific thinking helps you understand technology and **evaluate information critically**, even if you do not study science after Grade 10.
- **Textbook symbols:** magnifying glass = careful observation; compass = direction (right models, right questions, knowing limits).

Quick revision: key terms

Model = simplified representation | **Law** = what happens | **Theory** = why it happens | **Principle** = broad guiding idea | **Prediction** = evidence-based expectation | **SI units** = common international units | **Estimation** = rough check of reasonableness

Pause and Ponder: think about these

1. Was a recent prediction (e.g., a cricket match result) based on evidence or guesswork? How can scientific thinking improve it?

2. Give one situation where an approximate answer is enough and one that needs an exact value.

3. Pick an object (pressure cooker, mobile phone) or problem (traffic jam) and sketch how at least two branches of science connect with it.

Activity 1.1: Model your bicycle ride from school to home. Which details do you keep, which can you ignore, and why is ignoring some useful?